

2MSSC Technical Program

Friday, September 30th, 2016

Session 1 - Hybrid and Real-time Hybrid simulation

Chair: Vedhus Hoskere

13:00-13:15 Frequency-Domain System Identification of a Multi-Actuator Loading Assembly for Multi-Axial Real-Time Hybrid Simulation Testing. **Gaston Fernandois Cornejo**

13:15-13:30 Displacement Estimation using Multimetric Data Fusion. **Jong-woong Park**

13:30-13:45 Parametric Identification of a Hydraulic Transfer System for Real-Time Hybrid Simulation. **Amin Magareh**

13:45-14:00 Modelling Error of Using a Shear Model to Model a Complex Building Structure. **Dongyu Zhang**

14:00-14:15 Parallel RTHS: CyberMech. **Johnny Condori**

14:15-14:30 A Benchmark Problem in RTHS. Multi-Degree-of-Freedom System. **Christian Silva**

14:30-14:45 Development of a Control System and User Interface for the Quanser Shake Table II. **Vivian Wong**

14:45-15:00 Multi-Stage Coil Magnetorheological Damper: Test, Model and Control. **Feihong Xu**

Session 2 - Computer Vision

Chair: Seungwook Seok

15:30-15:45 Automated Collection and Measurement for Construction Pay Items. **Chulmin Yeum**

15:45-16:00 Application of the Digital Image Correlation System Vic-2D. **Tu Hoang**

16:00-16:15 Computer Vision- Applications to Structural Health Monitoring. **Vedhus Hoskere**

16:15-16:30 Heterogeneous Sensor Networks for Effective and Efficient Multimetric Sensing. **Kirill Mechitov**

16:30-16:45 Vision Based Vibration Measurement. **Ziyi Zhao**

16:45-17:00 Automated Damage Detection in Miter Gates of Navigation Locks. **Brian Eick**

17:00-17:15 Autonomous Image Localization for Visual Inspection of Civil Infrastructure. **Jonseong Choi**

17:15-17:30 Introduction of Taisai Technologies and Projects. **Koji Tsuchimoto**

Saturday, October 1st, 2016

Session 3 - Applied Structural Dynamics

Chair: Johnny Condori

- 10:00-10:15** Free Vibration-Based System Identification through the Generation of Reliable Temporal Cross-Correlations. **Yasutaka Narazaki**
- 10:15-10:30** Reference-Free Displacement Estimation for Railroad Bridge Performance Evaluation. **Fernando Gomez**
- 10:30-10:45** Human-Structure Interaction Effects in Structures. **Daniel Gomez**
- 10:45-11:00** Higer Mode Effects of Tall Piers on Seismic Performance of Bridges. **Chen Xu**
- 11:00-11:15** Optimal Placement of Excitations for Stiffness Monitoring of Flexible Pretension Structures. **Xintao Wang**
- 11:15-11:30** The influence of Dynamic Parameters on Numerical Model. **Xiaoyu Liu**
- 11:30-11:45** State Control and Mechanical Analysis of the Self-anchored Suspension Bridge with Extra-wide Concrete Girder. **Zhou Guangpan**
- 11:45-12:00** Analysis of Damped Outrigger System. **Chuangjie Fang**

Session 4 - Structural Analysis, Design and Construction

Chair: Gastón Fernandois-Cornejo

- 13:00-13:15** Jointed Pipeline Characterization and Evaluation for Seismic Performance. **Addie Lederman**
- 13:15-13:30** Seismic Performance Evaluation for a Subway Station Structure by Incremental Dynamic Analysis. **Liu Tong**
- 13:30-13:45** Small-Scale Reconfigurable Frame for 6-DOF Shaking Table. **Amirali Najafi**
- 13:45-14:00** Rapid Repair Technology for Cement Concrete Pavement with Prefabricated Assembling Slab. **Zhang Min**
- 14:00-14:15** Shear Design Reinforcement Limits for High Strength Concrete Beams. **Sandra Villamizar**
- 14:15-14:30** Positive and Negative Shear Lag Behavior of Composite Twin-girder Decks with Varying Cross-section. **Li Zhu**
- 14:30-14:45** Physic-based Computational Model for Bond in Reinforced Concrete Structures. **Seungwook Seok**
- 14:45-15:00** Construction Process of Living Apartment in common city of China. **Lin Shurong**

Session 5 - Structural Health Monitoring

Chair: Chulmin Yeum

15:30-15:45 RTOS-based SHM System for Rare Event Detection and Rapid Condition Assessment.

Fu Yuguang

15:45-16:00 PZT Active Health Monitoring for Fatigue Accumulative Damage of Concrete Beam.

Maohua Zhang

16:00-16:15 Examination and Adaptation of the Data-Based Model-Free Hysteretic Restoring Force and Mass Identification Method. Alana Wilbee

16:15-16:30 Rapid Determination of Seismic Fragility Functions with Limited Observations.

Francisco Peña

16:30-16:45 Study of Filter Design and Data Processing Implementation in Xnode. Siang Zhou